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Preliminary evaluation of chemical component in the 2004 indian ocean giant tsunami impacted soil using a Co_2 Laser Induced Breakdown Spectroscopy (LIBS)

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Abstract. Preliminary evaluation of chemical component in soil impacted by 2004 Indian Ocean giant tsunami using LIBS has been carried out. The sampling of the tsunami impacted soil was carried out in several points in Aceh, Indonesia. In this study, the sample soil was pretreated to turn into pellets. The beam of a TEA CO_2 laser (10.6 µm, 3 J, 200 ns, 10 Hz) is sent onto surface of pellet using focusing lens under air at atmospheric pressure for producing a plasma. The emission spectral lines from plasma that generated were discovered when testing with Optical Multichannel Analyzer (OMA) system. It shows many emission lines because of various elements could clearly be detected and be identified. It was found that both the tsunami impacted and unimpacted soil sample contain chemical signatures of tsunami event such as saltwater signal (Na), carbonate markers (Ca, Mg, Sr), terestrial markers (Fe, Mn, Si Ti, Ba), and other heavy metal elements. However, it is found that emission intensities of the spectral lines are far higher for the tsunami impacted soil samples compared to the case of tsunami unimpacted soil. The qualitative observation finding implies that the chemical elements such as calcium (Ca), then followed by magnesium (Mg), and sodium (Na) are good candidate for chemical proxies of tsunami event. Since the measurement allows to simultaneously observe these elements in a single measurement, thus LIBS is a potential, simultaneous, efficient, and versatile spectroscopic technique not only for analyzing chemical component in soil affected by the tsunami but also to identify the tsunami impacted soil from the impacted soilThe first section in your paper.

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1. Introduction

The giant Ocean Indian tsunami that happened in 2004 have caused millions die of humans [1], destroying infrastructure, geology, and environment in long term condition including polution and contamination of chemical component such as, salt, heavy metals, and organic content in affected soils [2]. The wildness of this event was missed from the attention of coastal residents and calamitiy researchers. Damage and contamination resulted from tsunami shown that knowledge about tsunami was inadequate. Even though, specified knowledge regarding the giant tsunami both recent and prehistoric tsunami is weighty to be deeply studied for evaluating the vulnerability and preparedness, especially those living in coastal areas [3].

In fact, it has been found a high concentration of salt (Ca, Mg, Na, K, Cl, SO_4), and heavy metal content (Pb, Zn, As Cd, Cr, Ni, and Cu) in affected soil after tsunami in Thailand [2]. Then, similar phenomena occured in agricultural land in Banda Aceh city, namely the concentration of salt and metal component was found significantly elevated after tsunami disaster. The impacted soil was contaminated by heavy metals (Cu, Co, Fe, Zn, Cr, Mn, Cd, Pb, and Ni), and salt content (K, Ca, Mg, Li, Na) [4]. Therefore, in case of the environmental problems occurred after tsunami strike, contamination of tsunami in impacted soil regions have to periodically be evaluated. Moreover, monitoring contamination in impacted soil is important to examine and to extend the historical records of geological events, to provide conclusive evidence when other proxies are lost or vague, and to make a comparison to the past tsunami. Thus, the chemical signature is very useful to investigate the areas affected by a tsunami [5, 6, 7, 8, 9].

In general, analytical instrument that have been used for evaluation of chemical component in various sample namely Ion Chromatography (IC), Inductively Coupled Plasma (ICP), Flame Atomic Absorption Spectrometry (FAAS), and X-Ray Fluorescence (XRF). Nevertheless, these instruments shown tedious sample preparation, consuming a lot of samples, taking much time, draining energy, and not environmentally friendly. Considering such great affected areas, a fast, simultaneous, and effective analytical techniques that can make geochemical analysis of chemical component in soil affected by tsunami quickly and efficiently must be developed.

A new analytical instrument, namely Laser Induced Breakdown Spectroscopy (LIBS) could be detected and identified for almost all chemical elements in one measurement, easy to use, cheap, in situ, also could be used for analysis qualitative and quantitative [10], and it is able to analyze the various types of samples [11, 12]. Furthermore, LIBS is also sensitive tools to go towards for detecting an abundance elements in natural materials, such as Be, He, B, Li, N, C and O, where these elements are troubled to detect using other technique. LIBS has been used for geochemical analysis of various natural samples as minerals, rocks, soil, sediments, natural fluids, [13], analysis of salt, heavy metal pollution in soils [14, 15, 16, 17, 18]. Thus, owing to the advantages, LIBS is expected to be developed as a geochemical analysis tool for identification of tsunami affected soils in Aceh, the most affected region by 2004 Indian Ocean giant tsunami.

2. Method

The set-up of LIBS equipment was demonstrated in Figure 1. The implement of LIBS consisted of TEA-CO_2 laser with energy = 3, λ= 10.6 µm, 200 ns, f = 10 Hz, then optical detector, that is Optical Multichannel Analyzer (OMA). This OMA stand up by spectrograph (0.32 m focal length; 1200 graves/mm grating; and 1024 canal photodiode detector arrays) with a micro canal plate amplifier. In order to form plasma, a TEA-CO_2 laser beam was targeted onto the pellet superficies which fixed in coil metal expanse using lens of zinc selenium (ZnSe). The emissions from formed plasma were detected using OMA system. Then, this emission accumulated with optical fiber, and it is fed to spectograph usage slot. The optical signal from the spectograph detected using photodiode arrangement. The OMA system fixed on computer using SpectraView software. The resulted product were maintained in computer for further processing data. After that, the resulted spectrum was identified with comparing to database of NIST (National Institute of Standard and Technology). The

vicinity gas in chamber could be emptied with utilizing a pump, then it occupied with the desired gas at required pressure.

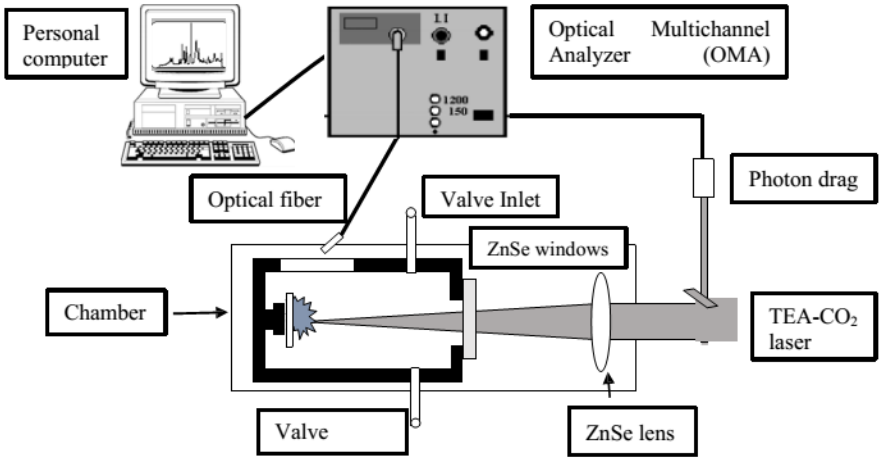


Figure 1. LIBS equipment using a TEA-CO_2 laser experimental setup.

As the preliminary research, all of soil samples were collected from several regions in Aceh, the most impacted area that happened by giant tsunami in 2004 such as Aceh Besar, Aceh Barat, and Banda Aceh regency. The soil was put and dried in the oven for 24 hours at 50°C to remove water content in the soil. Furthermore, the soil sample was crushed and mashed using a mortal and pestle. Then, the sample was sieved using a 100 mesh sieve to obtain particles with a homogeneous size. The soil powder from the sieve is compacted into a pellet form by weighing 5 grams and put into a pellet maker mold. Furthermore, it is put into the press machine and given a pressure force of 5 tons. After stand for 15 minutes, the pellet mold is removed from the press machine.

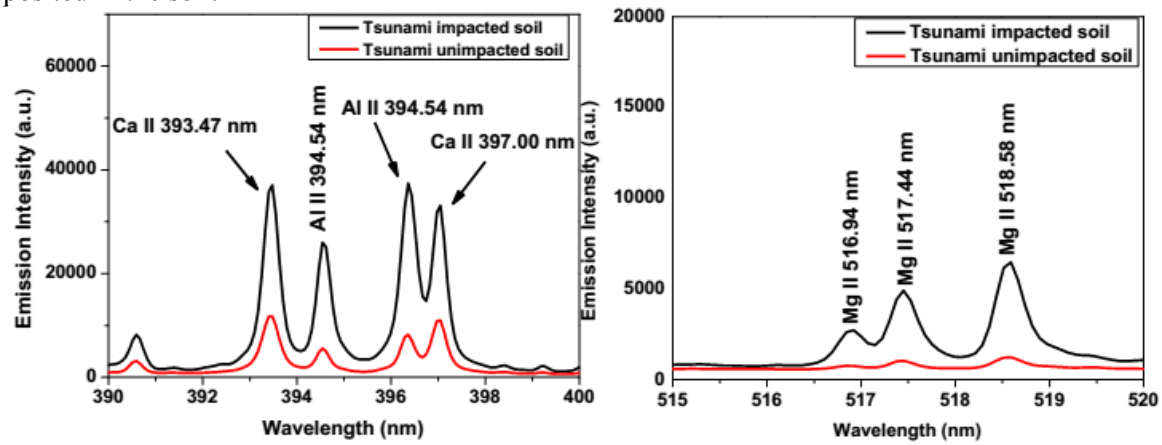
3. Result and Discussion

Aceh is the most impacted area when tsunami happened in 2004. After this event, the soil that impacted by tsunami was poluted and contaminated by seawater, tsunami sediment, and other materials that brought from tsunami wave. In case analysis soil sample using LIBS technique, the sample generally were made into pellet form to cover the surface sample having scatter, that could be identified the emission lines of the element was difficult. The emission spectrum could be observed from the plasma generated in form of pellets soil. The horizontal axis represents the wavelength (nm), while the vertical axis represents the emission (a.u.). According to [4], the agricultural land in Banda Aceh city which hit by tsunami contained chemical components including salt, metal and heavy metals, as well as organic elements in high concentrations. The chemical element of sample has many emission lines with different intensities based on a database published by the National Institute for Science and Technology (NIST). Based on the result, it shows that the spectrum in the 200 nm - 900 nm wavelength could be detected salt components (Ca, Al, Mg, K, Na), metals and heavy metals (Ti, Fe, Ba, Sr , Mn, Cr, Zn, Pb), and organic elements (H, O, and N).

Figure 2. shows the spectrum of generated plasma from soil pellet over a range 390 nm - 400 nm, that predicted the emission lines of Ca and Al theoretically appear. Spectral emission lines could be clearly seen when the scale of the wavelength is enlarged. Based on the results, the spectral atomic line of Ca (Ca II 393.47 nm, and Ca II 397.00 nm) appears very strong and clear intensity with respect to the background emission lines. The intensity emission of Ca II at 393.51 nm and 397.00 nm wavelengths in impacted soil is higher with values of 37080 and 33111 than unimpacted soil. Meanwhile, the low emission intensity of Ca was found in unimpacted soil, with intensity 11786 and 11011. Previous study by [19] reported that the affected soil by 2009 SPT tsunami on Samoa island contained Ca elements which indicated an indicator of coral reef material. Another study reported that tsunami in Mangaia contained Ca elements which indicated an indicator of sea water [20]. The high

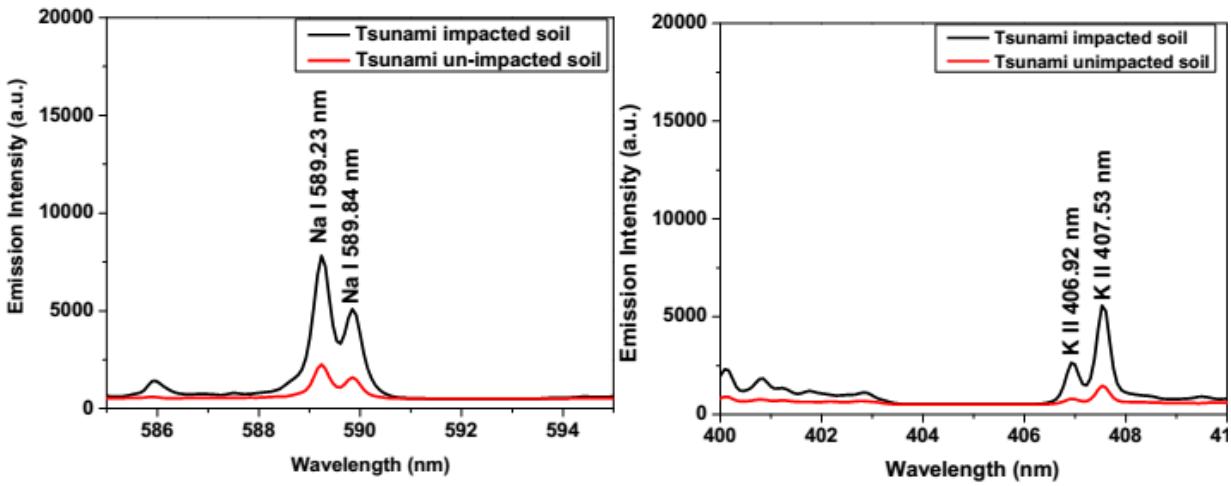
emission intensity of Ca elements in this study shown that the tsunami impacted soil contained carbonate material sourced from the sea such as shells and coral reefs deposited in soil.

Then, the Mg emission lines also appear very strong at range wavelength of 510 nm - 520 nm. Figure 2.b shows the Mg triplet atomic emission lines that detected at the wavelengths of Mg I 516.94 nm, Mg I 517.44 nm, and Mg I 518.68 nm. The higher intensity of Mg I emission at this wavelength was found at tsunami impacted soil with values of emission intensity of 12633, and 23594, and 26607, respectively. Meanwhile, the low emission intensity of Mg was found in tsunami unimpacted soil with values of 4195, 6452, and 8202 respevtively. The presence of Mg element in this study indicates that tsunami affected soil contained seawater, shell material, mixing of sea water with carbonic acid deposited in the soil.



(a) (b)

Figure 2. The atomic emission lines of chemical element from generated plasma from pellets in impacted and unimpacted soil samples at wavelength (a). 390 nm - 400 nm. (b). 515 nm – 520 nm.



(a) (b)

Figure 3. The atomic emission lines of chemical element from generated plasma from pellet in impacted and unimpacted soil samples at wavelength (a). 585 nm - 595 nm. (b). 400 nm – 410 nm.

Specific emission lines of Na I were identified and matched at 589.23 nm and 589.84 nm. Figure 3.a. shows the Na I atomic emission lines detected at wavelengths of 589.23 nm, and 589.99 nm. The emission intensity of Na element was found in higher concentration in tsunami impacted soil with a value of 7822 and 5093, while the emission intensitiy of unimpacted soil was 2265 and 1594. The presence of Na in the soil sample implies that the soil contains a saltwater signal, so that Na can be used as a chemical signature for a tsunami event. Then, after enlarging the wavelength range of 400

nm - 410 nm, the atomic emission lines K I were found at 406.92 nm and 407.53 nm (Figure 3.b.). The intensity emission of K at tsunami impacted soil found with intensity 5568, while tsunami unimpacted soil has value 1472. The presence of K in the soil affected by the tsunami indicates that it contains a salwater signal, so that K can be used as a chemical signature to mark a tsunami event. The emission lines of salt component Ca, Na, and Mg also clearly detected from tsunami impacted soil sample with high concentration using TEA CO_2 that published by [14]. Then, the high composition of salt content in the affected soil in Aceh area are similarly to those published previously by [2,4] in Banda Aceh city and Thailand impacted soil.

After having identification and evaluation using atomic spectra database, the emission spectra lines from heavy metal content (Fe, Mn, Ti, Cu, Cr, Ba, Co) have also been confirmed. The emission line from Cr is Cr atomic line, namely Cr I 302.24 nm. The emission line from Mn detected at wavelength Mn I 303.99 nm. In the same spectrum, the emission lines from Fe I were turned up at wavelength 304.44 nm, then 305.28 nm. In another spectrum, emission line from Ti I could be clearly detected at wavelength 323.26 nm, and the emission spectra line from Ba, Sr, and Cu confirmed at wavelength 455.16 nm, 460.62 nm, and 250.89 nm, respectively. However, the intensities of the Ba, Sr, and Cu spectral emission lines are very weak as compared to salts. The heavy metal components that appear on tsunami affected soil are in accordance with research conducted by [16]. The impacted soil in Banda Aceh city contained heavy metal elements such as Mn, Pb, and Fe. In addition, the impacted soil also contained amount of organic content H, N and O. The emission lines of organic element could be identified in this research, but the emission spectra lines are not shown in this manuscript. The plasma characteristics will be studied further and the ratio of analytes obtained from the impacted soil to the unimpacted will studied in detail to find a representative chemical signatures based on LIBS signal [5, 6, 7, 8, 9].

4. Conclusion

Analytical technique, that is LIBS was used for chemical signatures preliminary evaluation from tsunami affected soil collected from Aceh region, Indonesia, the most impacted area by the giant tsunami on 2004. The chemical signatures such as saltwater signal (Na), carbonate markers (Ca, Mg, Sr), terestrial markers (Fe, Mn, Si Ti, Ba), and other heavy metal could simultaneously be detected from the soil. The results shown that chemical signatures from affected soil could be evaluated using LIBS. Thus, LIBS is a potential, simultaneous, efficient, and versatile spectroscopic technique for analyzing chemical component in soil affected by tsunami disaster.

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